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BEHAVIORAL ECOLOGY –ORIGINAL RESEARCH

Not accounting for interindividual variability can mask habitat selection patterns: a case study on black bears

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Abstract Habitat selection studies conducted at the population scale commonly aim to describe general patterns that could improve our understanding of the limiting factors in species–habitat relationships. Researchers often consider interindividual variation in selection patterns to control for its effects and avoid pseudoreplication by using mixed-effect models that include individuals as random factors. Here, we highlight common pitfalls and possible misinterpretations of this strategy by describing habitat selection of 21 black bears *Ursus americanus*. We used Bayesian mixed-effect models and compared results obtained when using random intercept (i.e., population level) versus calculating individual coefficients for each independent variable (i.e., individual level). We then related interindividual variability to individual characteristics (i.e., age, sex, reproductive status, body condition) in a multivariate analysis. The assumption of comparable behavior among individuals was verified only in 40% of the cases in our seasonal best models. Indeed, we found strong and opposite responses among sampled bears and individual coefficients were linked to individual characteristics. For some covariates, contrasted responses canceled each other out at the population level. In other cases, interindividual variability was concealed by the composition of our sample, with the majority of the bears (e.g., old individuals and bears in good physical condition) driving the population response (e.g., selection of young forest cuts). Our results stress the need to consider interindividual variability to avoid misinterpretation and uninformative results, especially for a flexible and opportunistic species. This study helps to identify some ecological drivers of interindividual variability in bear habitat selection patterns.

Keywords Individual habitat selection pattern · Intrinsic characteristic · Multivariate analyses · Random slope coefficient

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Introduction

The study of animal behavior, both at the species and population levels, is a central concept in ecology, most especially for the study of relationships between wildlife and habitat (Rosenzweig 1981, 1991). Indeed, having a basic knowledge of their environment presents advantages to animals and allows them to adjust their use of different habitat types, based on intrinsic habitat characteristics (i.e., biotic and abiotic features), to efficiently meet their biological requirements (e.g., food acquisition, shelter and reproduction) (Jaenike and Holt 1991; Orians and Wittenberger 1991). This close relationship between wildlife and habitat explains why conservationists commonly use habitat selection to plan land management and achieve species recovery (Morris 2003; Klar et al. 2008).

Researchers commonly aim to describe general patterns that could improve our understanding of the most important limiting factors while only accounting for interindividual

variation to control its effect (e.g., Łomnicki 1999; DeCesare et al. 2012; Courbin et al. 2013). To do so, sampled animals are frequently selected from similar sex or age classes, thus limiting the scope of the study to a particular segment of the population. Furthermore, this decision does not remove other potential interindividual variations related to uncontrolled characteristics (Hebblewhite and Merrill 2008; Duchesne et al. 2010). Ignoring such interindividual variation in statistical analyses may yield uninformative results or conclusions that are only relevant to a segment of the population (Gillies et al. 2006). This segment often represents either the majority of individuals in the sample or individuals expressing an extreme directional response. This raises questions on the frequency of such issues in studies modeling behavior at the population level. Although complex to answer, we believe that this issue could be quite common when studying species that exhibit complex and plastic behaviors, especially if the sample is composed of animals of different sex, age, hierarchical and reproductive status or other important intrinsic characteristics.

As argued by Gillies et al. (2006) and Dingemanse and Dochtermann (2013), mixed-effect models are powerful tools to explore between- and within-individual variations. In habitat selection analyses, mixed-effect models have essentially been used to control for pseudoreplication associated with the use of several telemetry locations per individual. However, this approach relaxes the need to sample individuals that share similar characteristics (e.g., sex, age class) as long as the sample size is large enough to address the biological question of interest. Indeed, these models can incorporate varying responses to a given variable for both the slope and the intercept. Several other methodological developments have recently been proposed to integrate individual variation into habitat selection analyses (e.g., Duchesne et al. 2010; Fieberg et al. 2010; Northrup et al. 2015; Prokopenko et al. 2017). Such methodological advances provide an interesting opportunity to link individual variability in habitat selection patterns to an individual's intrinsic characteristics (e.g., fitness surrogates, age and sex). It could allow us to identify groups of individuals that behave similarly and, ultimately, explain why such variability occurs. Besides acknowledging and accounting for such variability, mixed-effect models contribute to the improvement of our understanding of population ecology. Although the amount of data needed is greater, we believe that such use of mixed-effect models outweighs the risk of missing (or misinterpreting) contrasting behaviors at the individual level.

These concerns apply especially to omnivorous and generalist species. Their ability to use a wide range of food sources and habitats allows them to exhibit efficient behavioral plasticity when faced with variable resource availability and landscape configurations. Ignoring individual variability for such species could increase the risk of obtaining misleading or incomplete results if opposite or largely different behaviors are expected. We used the black bear *Ursus americanus* to study interindividual variability as it is affected by different limiting factors depending on individual status (Elowe and Dodge 1989) and is adaptable to diverse habitats and food sources (Mitchell and Powell 2003; Bastille-Rousseau et al. 2011; Latham et al. 2011).

Previous attempts to describe black bear habitat selection yielded poor predictive results (Mosnier et al. 2008) or underlined variable responses (Bastille-Rousseau et al. 2011; Latham et al. 2011). We, therefore, believe that a successful description of habitat selection at the population level requires the explicit incorporation of individual variations related to intrinsic characteristics (e.g., age, body condition and reproductive status). We used home-range scale (3rd order; Johnson 1980) monitoring and Bayesian mixed-effect models to describe habitat selection patterns at the population level (i.e., individuals distinguished by a random factor yielding individual intercepts, but one slope for each variable for the whole population). We then compared these results with those obtained when considering interindividual variations (i.e., individual estimates for each independent variable and intercept) and related these responses to a set of intrinsic characteristics.

We hypothesized that averaging over individual differences may hide important effects that strongly vary among individuals because of their age, body condition and reproductive status. We predicted that strong effects might be found for some individuals even if the population-level inference indicates little effect, as some individual characteristics could help to distinguish strong opposite responses between bears. For example, we predicted that refuge habitats (e.g., mature forest) would be selected for by females with cubs-of-the-year, as they tend to be risk-averse, while dominant males or individuals with poor body condition were expected to avoid such closed canopy land cover classes (Rogers 1987). We predicted that refuges would not be as important for these segments of the population, because they are less susceptible to predation and are, therefore, less vulnerable in more open environments. Furthermore, lone females will use open habitats more intensively to take advantage of the abundant and energetic food sources (e.g., berries) that they offer and that are so important for bears to accumulate fat reserves during their short active season (Eagle and Pelton 1983), even if these items are sometimes concentrated in risky environments (Young and Beecham 1986).

Methods

Study area

The study area was located north of Saguenay, Québec, Canada, and covered approximately 6 300 km^2 centered on Portneuf Lake (48°42′–49°17′N, 70°03′–70°42′W). Forests are transitional between the spruce *Picea* spp.—moss domain and the balsam fir *Abies balsamea*—white birch *Betula papyrifera* domain, and ~35% of the forested area was harvested during the last decades due to intense commercial logging. A dense coverage of deciduous trees regenerated with an abundance of willow *Salix* spp. and trembling aspen *Populus tremuloides*. We found a diversity of shrub species that produce berries; the most common were blueberry *Vaccinium angustifolium*; *V. myrtilloides*, serviceberry *Amelanchier* spp., bunchberry *Cornus canadensis*, skunk currant *Ribes glandulosum*, raspberry *Rubus idaeus*, wild and bristly sarsaparilla *Aralia nudicaulis* and *hispida* and chokecherry *Prunus virginiana*. In our study area, prey animals available to black bears were mainly beavers *Castor canadensis*, moose *Alces americanus* and caribou *Rangifer tarandus caribou* calves, snowshoe hares *Lepus americanus*, ruffed grouse *Bonasa umbellus*, spruce grouse *Falcipennis canadensis* and Canada geese *Branta canadensis*. Anthropogenic infrastructure is ubiquitous with a cabin density reaching 0.32 cabins/km^2 and forest road density of 1.8 km/km^2. The mean annual temperature ranges between −2.5 and 0 °C, and annual precipitation fluctuates between 1000 and 1300 mm, of which 30–35% falls as snow. The elevation ranges between 300 and 800 m with low rolling hills. In this study area, bear densities were estimated to range between 1.5 and 4.3 bears/10 km^2 using DNA mark–recapture based on hair-snagging (Chicoine 2014).

Telemetry survey

Bears were captured in June and July 2011 and 2012 using a padded foot snare or a culvert trap; then they were immobilized with a Telazol–Ketamine–Xylazine (5:4:1) mix; only adults (females >54 kg and males >68 kg) were equipped with GPS/Argos collars (TGW-4583H-2, Telonics, AZ, USA). All manipulations were approved by the Animal Welfare Committees of the Ministère des Forêts, de la Faune et des Parcs du Québec and of the Université du Québec à Rimouski (certificate #2011-30). Information and samples such as weight, length, a premolar tooth (for age determination) and the presence of young (visible cub, lactating cues or information collected during den visits in winter 2012) were collected to build an individual set of intrinsic characteristics. The 24 collars were programmed to record GPS locations at a 2-h fix-rate interval.

Statistical analysis

We used 1: 20,000 numerical forest maps updated in 2010 from the Ministère des Forêts, de la Faune et des Parcs du Québec, which provide information such as stand age, composition, height and density. Minimum resolution was 4 ha for forested area and 2 ha for non-forested areas (e.g., water bodies). We converted the vector maps into a raster of 10 × 10-m grain using ArcGIS 10.2.2. We classified land-cover types based on forest stand age and composition, and obtained the following categories: Lake, Swamp, Conifer, Cut (0–5 years old), Cut (6–20 years old), Regeneration (21–50 years old), Open, Mixed forest (51–120 years old), Mature forest (51–80 years old), Old growth forest (81–120 years old) (see Table 1 for a complete description).

We assessed habitat selection (Manly et al. 2002; Boyce et al. 2002) by contrasting land-cover types at bear GPS locations (use; coded 1) and at the same number of random points (availability; coded 0) distributed in each individual's seasonal home range (3rd order habitat selection; Johnson 1980). Individual home ranges were delineated using 100% MCPs (Mohr 1947) for three seasons (spring: den emergence to June 14th, summer: June 15th to July 31st, fall: August 1st to den entrance) differentiated by shifts in diet and in plant phenology (Lesmerises et al. 2015). Comparisons were

Table 1 Description of land-cover classes found under GPS locations (coded as 1, i.e., use) or under random points (coded as 0, i.e., availability)

Land-cover classes	Description
Cut (0–5)	Young forest cut of 0–5 years old
Cut (6–20)	Forest cut of 6–20 years old
Regeneration	Disturbance (natural or anthropogenic) of 20–40 years old
Open	Open non-regenerated disturbance of more than 20 years old
Mixed	Mixed deciduous and coniferous stand older than 50 years old
Mature conifer	Coniferous stand of 50–80 years old
Old conifer	Coniferous stand older than 81 years old
Swamp	Swamp and alder stand
Lake	Lakes, ponds and large rivers
Other	Other sparsely distributed land-cover classes

made using binomial generalized linear mixed models. The reference habitat type was "Regeneration", a largely represented land-cover type used proportionally to its availability by most of our collared bears. We preferred to use a Bayesian analytical framework, suggested by Hadfield et al. (2010), available in the package {MCMCglmm} (Hadfield 2010) in R 2.15.3 (R Development Core Team 2013), as our goal was to extract conditional responses (sensu Gillies et al. 2006). Moreover, our choice of method was driven by the fact that frequentist approaches fail to correctly estimate error around individual estimates (commonly called best linear unbiased predictors, BLUPs; Hadfield et al. 2010). Bayesian GLMM allows the inclusion of random factors while yielding credible intervals efficient at handling non-normal error distributions based on Monte Carlo Markov Chain iterations. We specified prior for fixed effects to be normally distributed and diffuse with mean = 0 and a large variance (10^8). As recommended by Hadfield (2010) for binary-response GLMMs, the residual variance was fixed at 1. We used an inverse Wishart distributed prior for random effects and set variance = 1 at the limit with low belief parameter (nu) = 0.002. The low nu value was used to allow the prior to be mainly uninformative. Models were run for 200,000 iterations with a burn-in discarding of 30,000 and a thinning interval of 50 to keep autocorrelation between thinned samples at a low level (<0.10) while ensuring large independent effective sample sizes of 1600–3400 for each term in the models. We visually examined MCMC chain convergence and mixing properties as well as density plots for every model parameter.

The analysis was divided into two parts, the first one being the population level, which aimed at determining if adding random slopes yielded a better model fit than only including a random intercept and determining if the model with random slopes was more parsimonious. The second part of the analysis is the individual level, which assessed if the variables that were identified as relevant at the population level had the same value for each individual bear and related individual characteristics to individual responses (i.e., coefficients) in a multivariate analytical framework to identify ecological drivers of variability in behavior. This two-step analysis was preferred to the approach advocated in Dingemanse and Dochtermann (2013) where intra- and interindividual variation was assessed directly in the model and was related to other variables (e.g., sex, age). We did so because our limited sample size prevented us from using interaction terms between each habitat and all intrinsic characteristics. Indeed, for each habitat, five intrinsic characteristics needed to be included as interactions, meaning the addition of 45 parameters in our habitat selection model. Then, we decided to extract conditional (i.e., individual) coefficient estimates and link them to the intrinsic characteristics in a subsequent analysis (see Benson et al. 2015 for a similar analytical approach).

We built two seasonal candidate models to account for the bears' changing behaviors and nutritional needs during their active period (Table 2). Both candidate models included all land-cover types except regeneration stands (used as the reference category in our habitat selection analysis, since it was used proportionally to its availability, as stated above, according to a priori descriptive statistics). In addition to land-cover classes, the first model included a random intercept while the second model generated individual slopes for each land-cover class while still calculating a random intercept. These two models were compared using the deviance information criterion (DIC) (Table 2), an information-theoretic model comparison criterion similar to AIC but designed for Bayesian regressions (Spiegelhalter et al. 2002; Hooten and Hobbs 2015). We evaluated the fit of the models using a k-fold cross-validation based on 50 iterations, using a training set containing 75% of the original dataset. The remaining 25% was used to assess model performance in discriminating presence and availability (Boyce et al. 2002). Koper and Manseau (2009) recommended removing a proportion of individuals instead of a proportion of the data from all individuals to build the training dataset and to obtain marginal (population-level) predictions (as commonly used based on Boyce et al. 2002). However, we preferred to remove a portion of the dataset for each individual to build the training dataset, as random slopes are determined on an individual basis and because model performance with random slopes cannot be assessed if individuals in the testing dataset are not all present in the training dataset. We acknowledge that this is a non-negligible difference between these two validation procedures

Table 2 Comparison of the mean adjustment (±SD) of the model when considering random intercept only or random intercept and random slopes, based on 50 iterations of k-fold cross-validations and differences in deviance information criterion (ΔDIC)

Model	Spring			Summer			Fall		
	K-fold cross-validation		ΔDIC	K-fold cross-validation		ΔDIC	K-fold cross-validation		ΔDIC
	Mean	SD		Mean	SD		Mean	SD	
With random intercept only	0.92	0.04	865	0.75	0.1	843	0.92	0.05	1659
With random intercept and random slopes	0.96	0.02	0	0.96	0.03	0	0.99	0.01	0

and that the latter approach yields inflated measures of fit through conditional predictions. However, we only used this validation to compare the relative performance of our models when considering or not random slopes.

We linked differences in behavioral responses to environmental variables (i.e., sets of regression coefficients for each individual bear) to intrinsic variables, i.e., age, weight (a proxy of the hierarchical status; Rogers 1987), reproductive status (female with a cub, with a yearling, or lone female, male) and body condition index (BCI; based on weight and total length, see Cattet et al. 2002). We did so using a redundancy analysis (RDA, package {vegan}; Oksanen et al. 2013) to compare the matrix of individual environmental coefficients to the matrix of intrinsic variables.

We standardized all variables (z-score) because of differences in orders of magnitude between them. We ensured that no multicollinearity occurred among explanatory variables (variance inflation factor <3; Graham 2003) and that no collinearity ($r < 0.6$) occurred between dependent variables; we then removed the weight variable because of collinearity with age and sex. We finally conducted an ANOVA on the RDA axes, followed by a permutation test (1000 iterations) to establish vector length, direction and significance of environmental variables.

Results

Telemetry survey

GPS locations from 18 bears (spring; $\bar{x} = 542$ fixes $\pm$ 65 (SD), 21 bears (summer; $\bar{x} = 444 \pm 110$) and 20 bears (fall; $\bar{x} = 704 \pm 179$) were recorded and used for the analyses.

Population level

Adding random terms increased the fit of the models, as models including random intercept and random slopes yield a lower DIC value and a higher k-fold cross-validation *r*-value (ranging from 0.96 to 0.99 depending on the season; Table 2). During spring, the coefficients of four variables had their credible interval excluding 0, as Cut (6–20) and Lake were avoided while Mixed and Old conifer were selected (Table 3). In summer, all variables having an influence on bear relative probability of occurrence were avoided (i.e., Mature conifer, Swamp and Lake). The pattern was the same during fall, except that another variable had an effect [i.e., Cut (0–5) were selected; Table 3].

Individual level

We extracted individual estimates for each variable for each season (Figs. 1, 2, 3) and highlighted opposite responses (i.e., selection and avoidance) among sampled individuals for five variables in spring and summer, and for six variables in the fall. Conversely, only one variable (i.e., Lake) was found to trigger the same response (i.e., avoidance) for all individual bears.

The first axis of the RDA was significant for all seasons and the second axis was significant in summer (Fig. 4). The proportion of the variance explained by the first two axes varied between 45 and 55% depending on seasons. Although direct comparison is not possible with the fit measures of population-level models, the variance explained is very satisfactory (Fig. 1), considering the small number of variables used in these analyses (Økland 1999). Selection patterns toward some habitat variables varied among individuals. In spring, females with cubs selected Old conifer more intensely, while males and bears in better body condition avoided Cut (6–20) less and used them proportionally to

Table 3 Description of most parsimonious population model for each season (spring: May 15th–June 14th, summer: June 15th–July 31st, fall: August 1st–September 14th)

Variable	Spring			Summer			Fall		
	β	L-95% CI	U-95% CI	β	L-95% CI	U-95% CI	β	L-95% CI	U-95% CI
Cut (0–5)	−0.72	−1.40	0.04	−0.22	−1.27	0.76	**1.19**	**0.60**	**1.89**
Cut (6–20)	**−0.37**	**−0.76**	**−0.01**	0.13	−0.26	0.54	0.09	−0.63	0.81
Open	0.10	−0.93	1.19	0.37	−0.35	1.09	0.29	−0.26	0.78
Mixed	**0.67**	**0.06**	**1.19**	0.41	−0.08	0.94	0.03	−0.71	0.70
Mature conifer	−0.19	−0.86	0.44	**−0.83**	**−1.62**	**−0.10**	**−0.66**	**−1.19**	**−0.11**
Old conifer	**0.71**	**0.38**	**1.02**	−0.04	−0.39	0.28	−0.02	−0.59	0.55
Swamp	−0.36	−0.98	0.16	**−0.63**	**−1.15**	**−0.20**	**−0.87**	**−1.79**	**−0.01**
Lake	**−3.04**	**−3.86**	**−2.16**	**−3.50**	**−4.23**	**−2.84**	**−3.14**	**−3.74**	**−2.60**
Other	0.06	−1.18	1.16	0.68	−0.26	1.60	0.55	−0.48	1.56

Coefficients (β) that have a 95% confidence interval (95% CI) that did not include zero are in bold font

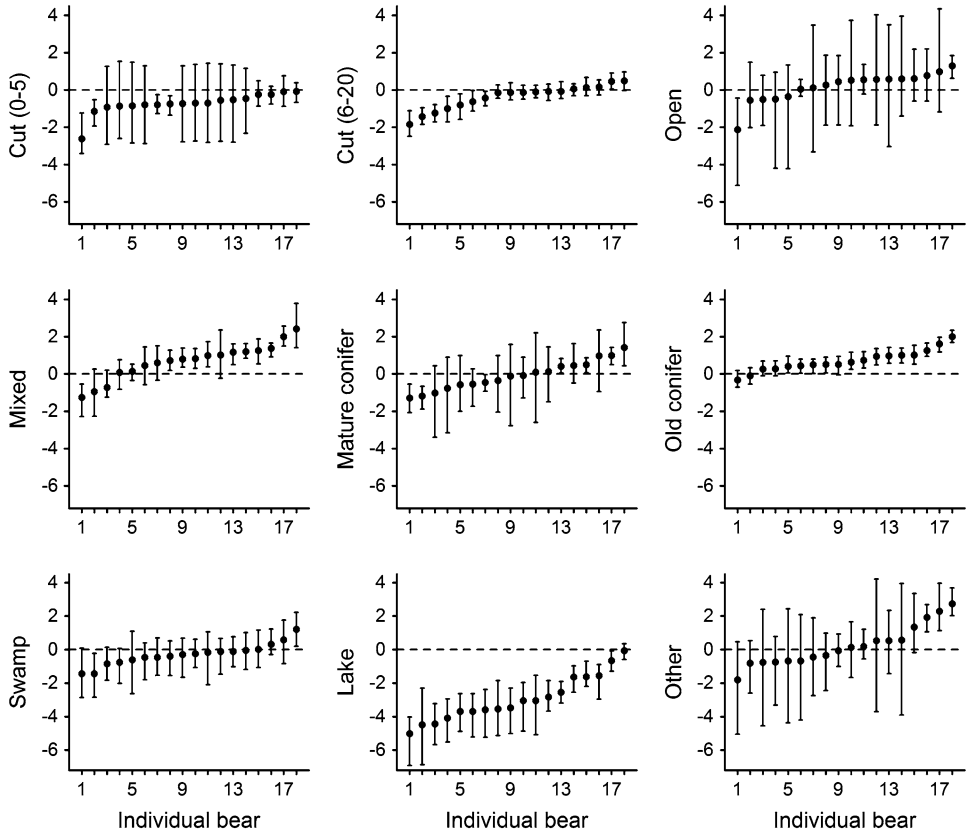

Fig. 1 Individual posterior mode and credible interval (95%) for each habitat in spring. Bears were ranked in increasing order of coefficient for each land-cover class

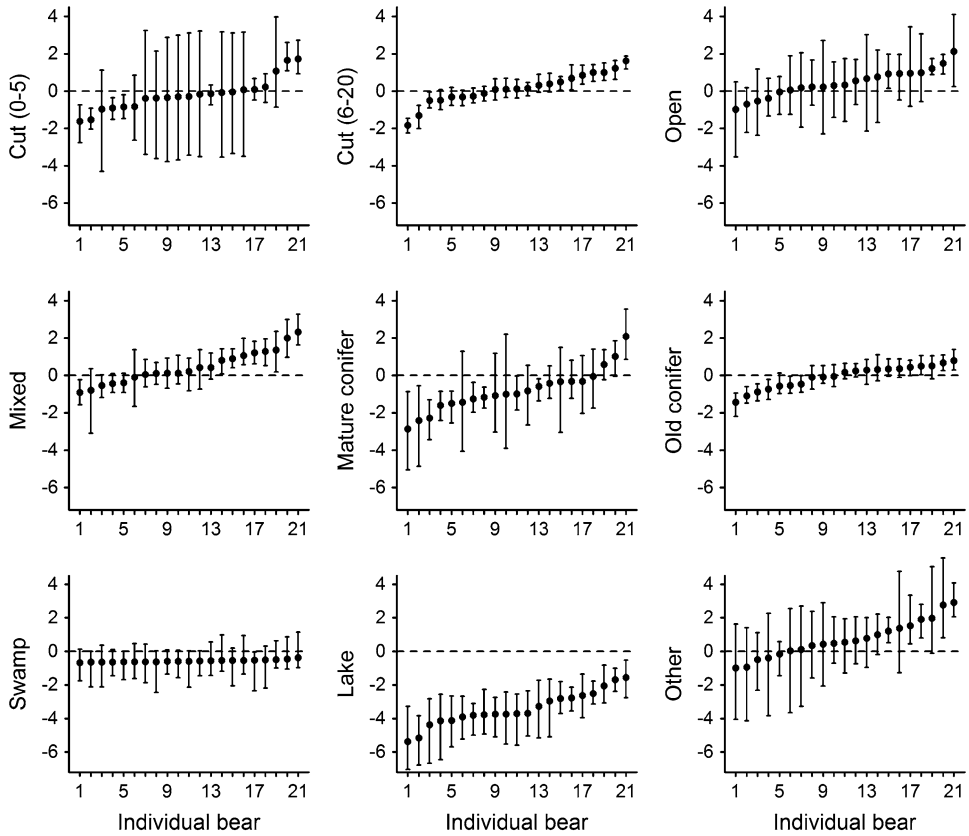

Fig. 2 Individual posterior mode and credible interval (95%) for each habitat in summer. Bears were ranked in increasing order of coefficient for each land-cover class

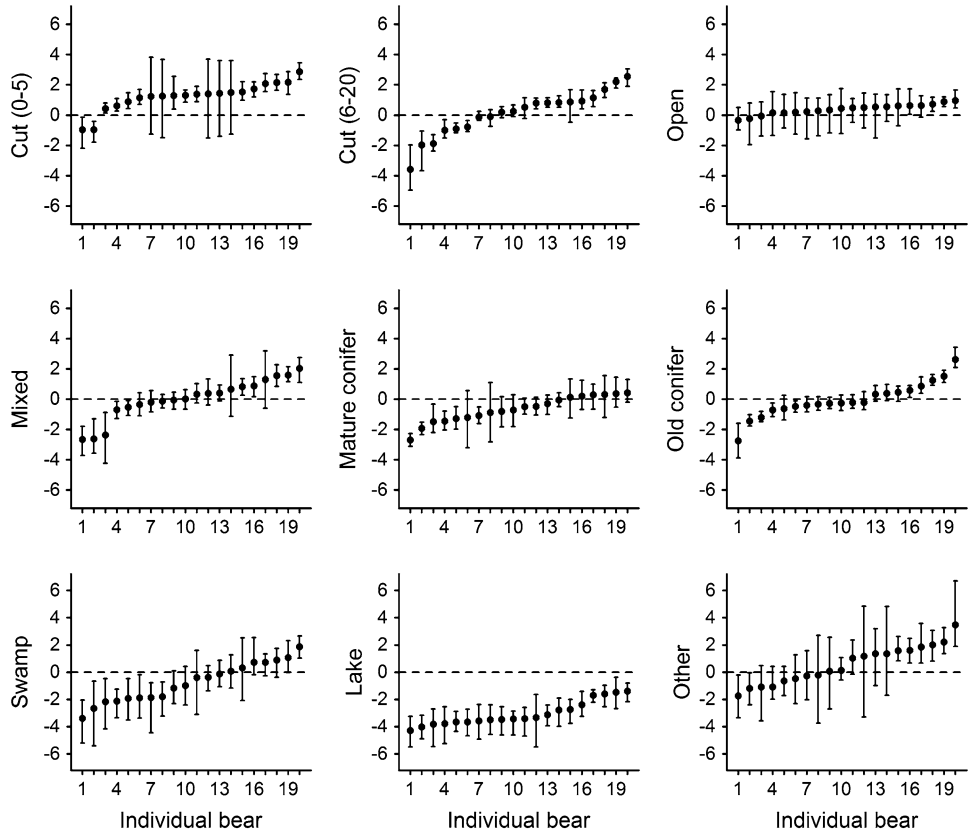

Fig. 3 Individual posterior mode and credible interval (95%) for each habitat in fall. Bears were ranked in increasing order of coefficient for each land-cover class

their availability. During summer, older individuals and, to a lesser extent, bears in better body condition seemed to use Cut (0–5) more than other individuals, especially lone females. In fall, lone females and males showed opposite responses to Mixed and Swamp land-cover classes, as the lone females selected mixed stands while males avoided them and selected wetlands. Older bears and individuals in good physical condition selected Cut (0–5) more strongly.

Discussion

Interindividual variability reveals contrasted habitat selection patterns

According to our hypothesis, our population-level models poorly explained bear habitat selection, especially in summer, when using only one response pattern for each independent variable. In contrast, allowing all individual bears to respond contrastingly led to a better fit, as demonstrated by the k-fold cross-validation. Furthermore, when linking individual estimates to sex, reproductive status, age and body condition, a high proportion of variance was explained and such relationships lead to a more complete understanding of habitat selection behaviors. Our results support our prediction, as opposite individual effects had been observed even if population-level inferences suggested little effect. For example, mixed stands were used proportionally to their availability at the population level during fall (i.e., the 95% CI of the regression coefficient overlapped zero), but when dissecting variability among individuals, we showed that males had opposite responses to those of lone females (see Fig. 4). As these two groups included almost the same number of individuals, their contrasting responses led to an inconclusive effect on habitat selection patterns at the population scale, as they canceled each other out. This phenomenon was not trivial, as it can be applied to more than 40% of the non-significant variables of our seasonal models.

Similarly, strong and clear responses observed at the population level could also conceal inherent interindividual variability. For instance, the explicit selection toward Cut (0–5) in fall at the population level was not as clear when considering individual responses (see Fig. 3). Moreover, when related to individual characteristics, we observed that older bears in good physical condition were more prone to select this land-cover class (see Fig. 4). As most bears selected young forest cuts, the population-level estimate was significant, but it hindered contrasting behavior, probably from young and individuals in poor physical condition. Conversely, no variability in bear response to lakes was noticed during all seasons in both analyses, as all bears avoided lakes. We found a similar pattern (no variation between

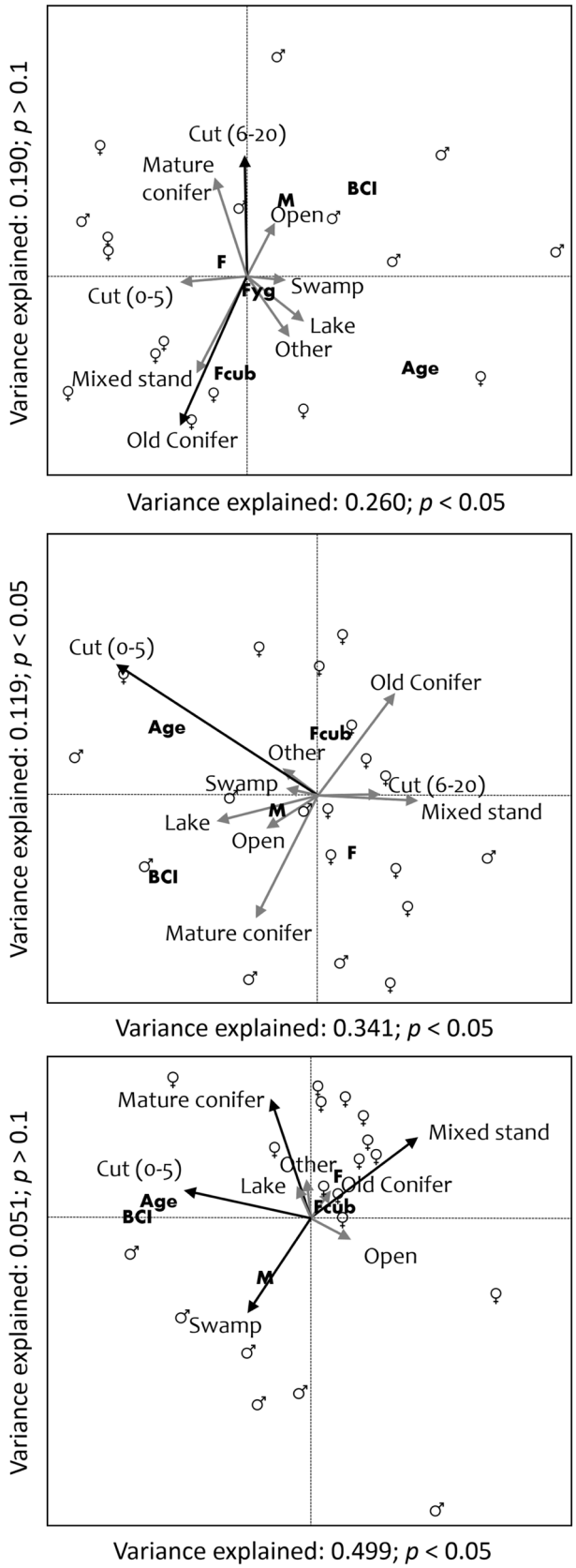

Fig. 4 Graphical representation of the RDA analyses using individual coefficients obtained from random interaction between individuals and each variable of the season's best models shown in Table 3. Each panel represents a season: **a** spring (May 15th–June 14th), **b** summer (June 15th–July 31st) and **c** fall (August 1st–September 14th). *Arrows* in *black* are significant ($p < 0.05$) while *arrows* in *gray* are not ($p > 0.05$). Individual characteristics include: *BCI* body condition index, Age, *M* male, *F* female without a cub, *Fcub* female with a cub-of-the-year and *Fyg* female with a yearling. *Gender symbols* represent the sex and the relative position of individuals

bears) for 5 out of 10 significant variables. Moreover, among the five land-cover types for which no behavioral responses could be linked to individual characteristics, one more displayed contrasting pattern among bears (i.e., mature conifer in summer, Fig. 2), suggesting that other factors could trigger such variability.

Ecological drivers of behavioral variability in bears

Some of the previous examples referred to sex-related behavioral differences, an intuitive and common concern which is usually addressed in habitat selection studies on bears by considering each sex separately (e.g., Mosnier et al. 2008). Nonetheless, other individual characteristics had important influence on black bear habitat selection patterns and could act as ecological drivers, but their effect was not apparent when individuals were all pooled in mixed-effect population models. Even using a small sample size, our study revealed interindividual variability in habitat selection that was related to bear reproductive status, age, sex and body condition, highlighting the contrasted resource requirements and compromises made by bears between food acquisition and refuge from predation (intra- and interspecific, as well as human avoidance; LeCount 1987; Farley and Robbins 1995; Lesmerises et al. 2015).

In spring, we showed that females with cubs selected old conifer forests, in contrast to males. Mature stands provide cover and refuge for cubs, known as good climbers, as they can access a large tree to climb and hide in on a shorter distance than in cutblocks (Lesmerises and St-Laurent, unpublished data). Rogers (1987) had already shown that females with cubs avoid areas where encounter risks with males are higher, especially when cubs are less mobile and during the mating season, when males are seeking for sexually receptive females and are then more prone to infanticide (LeCount 1987). In summer, older individuals in good body condition selected young cutblocks (Cut 0–5) more intensely, a land-cover class in which they find diversified and abundant food sources (Mosnier et al. 2008) such as wood-dwelling ant colonies and berries (Noyce et al. 1997; Lesmerises et al. 2015). A similar pattern was observed during fall for females that were selecting mature conifers and mixed stands, while

males were selecting swamps. During fall, a companion study showed that bears can find berries, grasses and willows in swamps (Lesmerises et al. 2015), i.e., important food resources during hyperphagia to store enough energy reserves before hibernation (Powell et al. 1997). Dominance of males over females could have led to the displacement of females from food-rich habitats to suboptimal—but safer—habitats. This "displacement hypothesis" is also supported by a stronger selection of young Cut blocks (Cut 0–5) by older individuals in good body condition. Females must, therefore, cope with poorer habitats to acquire enough resources to compensate for the energetic demand of lactation during previous summer or of embryo development during the upcoming winter (Farley and Robbins 1995). Doing so, they trade food availability and quality for more safety for their cubs.

Other drivers of interindividual variability in behavior

In addition to the individual characteristics considered in our analyses, we recognize that other intrinsic factors such as parental care or learning (Nielsen et al. 2013), behavioral plasticity (Careau et al. 2008; Duckworth 2010), biotic and abiotic developmental condition (Feil and Fraga 2012; Sweeney et al. 2013) and social interactions (Bergmüller and Taborsky 2010) could explain interindividual variability in habitat selection patterns and ultimately shape behavior; however, these are almost impossible to assess without exhaustive longitudinal surveys. Regardless of the cause, differences in behavioral responses to habitat heterogeneity are interesting, especially when strategies can be related to fitness or to evolutionary processes, and are part of the increasingly studied concept of personality (Dingemanse et al. 2002; Dochtermann and Dingemanse 2013). Moreover, a better linkage between behavior and conservation is needed to produce theory-based science using the latest available knowledge, leading to reliable and applied outcomes able to sustain species management and recovery (Angeloni et al. 2008).

It could be argued that contrasting behavioral variations between individuals essentially equates to describing the statistical "noise" around the average habitat selection patterns of a population or species. According to this, studies carried out at the population level, while yielding insightful average individual responses, could be more appropriate to identify limiting factors and orient management actions to ensure the survival of most individuals and, ultimately, help population maintenance and recovery. Although such an argument could be verified in some cases, an increasing body of literature underlines that individuals that appear physically similar but express contrasting behavioral strategies may have different fitness values (e.g., Duckworth and Badyaev 2007; Smith and Blumstein 2008, 2010; Bergmüller and Taborsky 2010; Adriaenssens and Johnsson 2013). To link such strategies to fitness, individual behavior must be stable and consistent over time (i.e., a behavioral syndrome or personality; Sih et al. 2004, 2012), which influences food acquisition (Johnson and Sih 2007), social and species interaction (Sih et al. 2012; Wolf and Weissing 2012) and mating (Smith and Blumstein 2008; Pruitt and Riechert 2009). Such an assessment of repeatability in behavior has already been achieved by Leclerc et al. (2015) for habitat selection in brown bears *Ursus arctos*. Also, Fortin et al. (2008) assessed that different strategies observed in a population could be rewarding depending on population density and competition, leading to equilibrium of competing strategies. It suggests that habitat selection patterns described at the population level could yield different results depending on population size. This makes management plans based on such patterns strongly dependent of current population size and potentially ineffective following population declines.

Conclusions

Highlighting different habitat selection patterns in an animal population provides many interesting opportunities, as it can orient management strategies and recovery actions by identifying habitat features that are suitable for individuals belonging to a critical, sensitive and less abundant segment of the population. From a conservation point of view, it will allow researchers to pinpoint key habitats to preserve as well as sink habitats to restore. If preservation of interindividual variability is a concern, this approach will undoubtedly perform better than using a mean population-level response that can hide rare or under-sampled phenotypes. Conversely, control of pest or overabundant species will benefit from highly valuable cues such as the habitat features selected by the most productive individuals.

For black bears, our study has shown that many intrinsic factors such as age, sex, body condition and reproductive status could explain variation in habitat selection, and that we could reveal contrasted patterns even in a small sample of collared animals. Many different factors could explain interindividual differences in behavior, and more research efforts should be invested in disentangling their respective influence on the evolutionary potential of this fundamental behavior that links individuals to resources (Leclerc et al. 2015). Consequently, we think that researchers should gather as much information as possible on each sampled animal to relate individual response toward various habitat variables to individual characteristics. We believe this is an efficient approach that could lead to a more realistic understanding of interindividual variability in animal behavior.

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Author contribution statement Conceived and designed the experiments: R.L. M.H.S.L. Performed the experiments: R.L. MHSL. Analyzed the data: R.L. Contributed reagents/materials/analysis tools: M.H.S.L. Wrote the paper: R.L., M.H.S.L. Coordinated the funding of the project: M.H.S.L. Prepared, validated and submitted the capture protocol to the Canadian Council on Animal Care: M.H.S.L.

Compliance with ethical standards

Ethical approval Black bear captures were conducted by the technicians of the Ministère des Forêts, de la Faune et des Parcs du Québec (hereafter MFFP) in June and July 2011 and 2012. All manipulations were approved by the Animal Welfare Committees of the MFFP and of the Université du Québec à Rimouski (certificate #2011-30).

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Conflict of interest The authors declare that they have no conflict of interest.

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